

E-Commerce SQL Portfolio Project

Whole Project Insights Report — Customer Analytics Database

Project Overview

This project is a comprehensive SQL portfolio piece built around a real E-Commerce Sales & Customer Analytics dataset (Excel-sourced, imported into MySQL). It demonstrates end-to-end SQL proficiency — from schema design and data import, through basic querying, to advanced analytical techniques used in real-world business intelligence and data analytics roles.

Database: **Customer_Analytics_db**

Dataset size: **77,445 rows** across 8 relational tables (verified from source Excel workbook)

Total queries written: **150 SQL queries**, organized into 3 skill-based sections

Database Schema — 8 Tables (Actual Row Counts)

Table	Rows	Purpose
Categories	20	Product category reference data
Suppliers	150	Supplier information for sourced products
Employees	100	Staff records, departments, and reporting hierarchy
Customers	5,000	Customer profile & signup information
Products	2,000	Product catalog (pricing, stock, category, supplier)
Orders	15,000	Order-level transaction data (status, dates, shipping)
OrderItems	45,175	Line-item level detail for each order
Reviews	10,000	Customer product reviews and ratings

Total: 77,445 rows

Query Breakdown by Skill Level

Section	Range	Focus Areas
Basic Queries	1–55	SELECT statements, WHERE/IN/BETWEEN filtering, ORDER BY, LIMIT, aggregate functions
Intermediate Queries	56–105	INNER/LEFT/Self-Joins, HAVING, subqueries, CASE statements, date & string manipulation
Advanced Queries	106–150	Window functions (RANK, ROW_NUMBER, LAG, LEAD, NTILE, PERCENT_RANK), CTEs

Key Analytical Techniques Demonstrated

- Multi-table joins (INNER, LEFT, Self-Joins) for relational data analysis
- Window functions — RANK, DENSE_RANK, ROW_NUMBER, LAG, LEAD, NTILE, PERCENT_RANK
- Common Table Expressions (CTEs), including Recursive CTEs for employee/manager hierarchies

- Market Basket Analysis — identifying frequently co-purchased product pairs
- RFM Analysis (Recency, Frequency, Monetary) for customer segmentation
- Cohort-style customer churn and retention analysis
- SQL Views for reusable reporting layers (e.g., product performance, customer summary)
- Conditional aggregation / pivot-style reporting using CASE statements
- Correlated subqueries for row-level comparative analysis

Views Created

- **product_performance** — summarizes units sold and revenue by product
- **customer_summary** — summarizes each customer's order activity

Dataset Snapshot — Key Distributions

Orders by Status (15,000 total)

Order Status	Count	% of Orders
Delivered	6,795	45.3%
Shipped	2,988	19.9%
Processing	2,214	14.8%
Pending	1,492	10.0%
Cancelled	912	6.1%
Returned	599	4.0%

Other Highlights

Metric	Value
Product price range	\$5.38 – \$1,999.20
Active products	1,807 (90.4%)
Discontinued products	193 (9.6%)
Top customer country	USA — 1,200 customers
Review rating spread	~2,000 reviews per star (1–5), evenly distributed
Largest employee department	Management — 14 employees

Project Highlights

This project mirrors the kind of analysis a data analyst performs in an e-commerce business: tracking sales performance, understanding customer behavior, segmenting high-value customers, identifying churn risk, and building repeatable reporting views — all using progressively advanced SQL, from single-table filters to recursive hierarchies and RFM segmentation. The dataset's balanced review ratings and healthy delivery rate (45%) also make it well-suited for customer satisfaction and fulfillment-performance storytelling.

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